A Separable Response Covariance for the Covariance-Moderation DGP

2026-09-30 11:12 PDT; expanded 2026-10-01 11:46 PDT

Author. pmsimstats team

Revision 2026-10-07.

- **Added:**
  - an intuitive overview ("In brief");
  - the general ceiling with prognostic and baseline couplings (Section 4.8);
  - the power cost of the higher ceiling (Section 6);
  - the simulation evidence now available (Section 10).
- **Corrected:**
  - Hendrickson's published ρ is 0.8; the figures quoted for `orig` were at the package value 0.7. Both are now given.
  - The coupling pattern is written g_t, not ϕ_t, which was also used for the AR(1) step correlation.
  - "The summed response is AR(1)" holds when the component SDs are constant over visits.
  - The analysis model that the separable structure matches includes a random intercept.
  - The `orig` ceiling depends only on visit count and on-share for binary coupling.
- **Later the same day (consistency review):** a notation index was added and colliding symbols renamed (see there); the stress-test description of Section 10 was corrected; all references were verified and given DOIs.

Purpose. The project is converging on a single covariance-based data-generating process (DGP) to replace the two now in use, the package's Architecture B (`covar`) and the published Hendrickson et al. implementation [1] at commit `58b32a9` (`orig`). An AR(1) within-factor kernel has been chosen. This note sets out one further structural choice, that the correlation among the three response factors across visits be *separable*, and examines it in detail: what it asserts, why it is coherent where the current construct is not, how it relates to Hendrickson's construct, what it implies for the largest testable biomarker correlation, and what it would change across the compendium.

Contents

Notation index and glossary . . . 3
In brief: what separable AR(1) means . . . 4
1. Summary . . . 5
2. Review of the original note . . . 5
3. Intuition . . . 6
4. Derivations . . . 6
4.1 Setup and the feasibility condition . . . 6
4.2 The separable structure . . . 7
4.3 The factorized ceiling and its switch-cost form . . . 7
4.4 Why the current `covar` fails . . . 8
4.5 Hendrickson's construct as two valid terms . . . 9
4.6 The summed response and the analysis model . . . 9
4.7 One framework for all three . . . 9
4.8 The ceiling with prognostic and baseline couplings . . . 10
5. Advantages and disadvantages . . . 10
6. Effect on the largest testable c_{bm} . . . 11
7. Impact across the compendium . . . 12
8. Alternatives considered . . . 13
9. Recommendation and implementation . . . 13
10. Evidence status and limitations . . . 14
11. References . . . 15

List of Tables

1 Notation index for the separable response covariance note . . . 3
2 AR(1) correlation between visits by weeks apart at $\rho = 0.7$ . . . 4
3 Cost of each coupling-pattern step under the AR(1) predictor . . . 8
4 Comparison of the `orig`, `covar` and proposed response covariance constructs . . . 10
5 Ceilings without carryover under the current and separable AR(1) structures by path . . . 11
6 Simulation engines with their response covariance construct and parameters . . . 12
7 Engine, cross-factor parameters and consequence of the change for each paper . . . 12

Notation index and glossary

Added 2026-10-07. Notation follows `analysis/report/NOTATION.md`. Symbols marked *local* are defined in this note only. On 2026-10-07 the eigenvalue symbols were renamed from κ, λ to μ, ω (both collided with canonical symbols), the coupling pattern from u to g, the `orig` person- and occasion-level effects from P_c, $E_{c,t}$ to T_c, $O_{c,t}$, and the AR(1) innovation from ε_t to ξ_t.

Notation index.

Table 1: Notation index for the separable response covariance note

Symbol	Meaning	Status
Y_{it}	outcome (code `Sx`)	canonical
BL_i, σ_{BL}	baseline level and its SD	canonical; σ_{BL} local
TV, PB, BR	natural-course, placebo-belief and drug response components	canonical
B	biomarker	canonical
c_{bm}	covariance-moderation correlation of B with BR	canonical
ρ	AR(1) correlation per week (CS correlation for `orig`)	canonical
κ	SE ratio, model-based over empirical (Section 10 only)	canonical
t_{sd}, $t_{1/2}$	time since discontinuation; carryover half-life	canonical
n, w_t, d_t	visits per path, visit week, gap $w_t - w_{t-1}$	local
R, $\tilde{r}$	full correlation matrix; biomarker correlation vector	local
M	$3n \times 3n$ response block of R	local
K, A	3×3 factor correlation; $n \times n$ time kernel	local
c_1, $c_\times$	cross-factor correlation at the same and at different visits	local
I, J, e_3	identity, all-ones matrix, unit vector selecting BR	local
g, g_t	coupling pattern: 1 on drug, residual fraction off drug, 0 before exposure	local
c^*_{bm}	ceiling: largest c_{bm} with $R \succ 0$	local
R^2	squared multiple correlation of BR on TV, PB at one visit	local
ϕ_t	AR(1) step correlation ρ^{d_t}	local
ξ_t	standardized AR(1) innovation	local
L, L_A, L_K, $\mathbf{Z}$, $\mathbf{X}$	Cholesky factors; standard normal and response matrices	local
μ, q_K; ω, q_A	eigenvalue and eigenvector of K; of A	local
$\omega_{\min}(A)$, $\mu_{\min}(K_o)$	smallest eigenvalue of A; of K_o	local
Q, Ω	eigenvectors and diagonal eigenvalue matrix of A	local
n_{eff}	effective number of independent visits	local
K_a, K_b	factor matrices of the `covar` decomposition	local
K_p, K_o, T_c, $O_{c,t}$	person-level and occasion-level factor matrices and effects of `orig`	local
K_s, K_j, A_j	factor matrices and kernels of a sum of separable terms	local
A_{CS}, A_{AR}	compound-symmetry and AR(1) time kernels	local
s_t, S_t	factor SDs at visit t; summed response component	local
v_{TV}, v_{PB}, v_{BR}, c_{bl}	biomarker correlations with each factor over visits, and with BL	local
$c_{bm,TV}$	prognostic coupling of B to TV (Section 4.8)	local

Glossary.

- **`orig`.** Hendrickson et al.'s published construct at commit `58b32a9`: compound symmetry within factor, constant cross-factor correlation.
- **`covar`.** The package's current construct (Architecture B): AR(1) within factor, cross-factor correlation c_1 at the same visit and $c_\times \rho^{\text{gap}}$ otherwise.
- **Separable.** A covariance that factors as $K \otimes A$, a factor part times a time part.

- **Ceiling.** The largest testable c_{bm}, set by positive definiteness of R.
- **Coupling pattern.** The visit-by-visit multiplier of c_{bm} in the biomarker row; *step* (published) or *graded*.
- **Linear model of coregionalization.** A sum of separable terms $\sum_j K_j \otimes A_j$.

In brief: what separable AR(1) means

Added 2026-10-07: an intuitive description, ahead of the formal treatment.

The setting. Each participant has three hidden responses that move over the trial:

- the drug response (BR);
- the placebo, or expectancy, response (PB);
- the natural course (TV).

At any visit each one is a little above or below its expected trajectory. The question is how those deviations relate across visits, and across the three responses.

"AR(1)": memory that fades with time. If a participant's natural course is running better than expected this week, it will probably still be better than expected next week, less so in a month, and hardly at all by the end of the trial. With $\rho = 0.7$ per week, the correlation between two visits of the same response is:

Table 2: AR(1) correlation between visits by weeks apart at $\rho = 0.7$

Weeks apart	1	2	4	16
Correlation, 0.7^{weeks}	0.70	0.49	0.24	0.003

Each week's deviation is mostly the previous one carried forward and shrunk, plus something new. Nearby visits look alike; distant visits are nearly independent.

"Separable": one clock for everything. The three responses are linked to one another at a fixed strength, a correlation of $c_1 = 0.2$ at the same visit. Separability says that this link fades with time on exactly the same clock as each response's own memory. The correlation between two responses at two visits is therefore the product of two pieces:

$$\text{Cor}(\text{response } c \text{ at week } t,\ \text{response } c' \text{ at week } s) = \underbrace{K_{cc'}}_{\text{which responses}} \times \underbrace{\rho^{|t-s|}}_{\text{how far apart}} .$$

For example, placebo at week 8 and natural course at week 12 are correlated $0.2 \times 0.7^4 \approx 0.05$. The "which responses" part and the "how far apart" part never interact.

The generative picture. Imagine three independent random processes, each drifting slowly with the same 0.7-per-week memory, and make each response a fixed blend of the three. Every correlation the model claims is then produced by something. That is why the correlation matrix can never be invalid, whatever the visit schedule (Section 4.2).

How it compares with the alternatives.

- **Hendrickson et al.'s construct (`orig`, compound symmetry).** Each person carries a permanent offset for each response, plus independent noise at each visit. Any two visits of a response are correlated at the published $\rho = 0.8$, whether one week or twenty weeks apart. The construct is valid, but its memory never fades. Sections 3 and 4.5 quote its implied quantities at the package value $\rho = 0.7$.
- **The package's current AR(1) form (`covar`, not separable).** It has fading memory, but it also asserts an extra same-visit link between responses that no underlying process generates. On closely spaced visits it asks for correlations no valid covariance can deliver, and the matrix fails (Section 4.4).

- **Separable AR(1) (proposed).** Fading memory on one clock, with nothing extra, so it is always valid.

Why choose it.

- **It is valid for any visit schedule.**
- **The largest testable biomarker effect has an exact formula.** Each on/off switch between nearby visits has a cost, so schedules with many switches can carry only smaller effects (Section 4.3).
- **It matches the analysis model.** The outcome also contains each person's permanent baseline level. The data therefore look like a random intercept plus fading AR(1) noise, which is what the random-intercept-plus-`corCAR1` analysis assumes. The match is close but not exact, because the placebo response's SD scales with expectancy (Section 4.6).

What it gives up.

- **All three responses forget at the same rate.** A faster-fading placebo response, for example, is not allowed (Section 8 gives the valid extension).
- **The responses carry almost no long-term memory.** Week 4 and week 20 are essentially unrelated within a response (correlation 0.003). Lasting person-to-person differences come only from the baseline level.

1. Summary

1. **The proposal.** Replace the response block of the correlation matrix with a Kronecker product $M = K \otimes A$: a 3×3 factor correlation K times a single AR(1) kernel A in calendar time. Lagged cross-factor correlation becomes $c_1 \rho^{\text{gap}}$, and the parameter `c.cfct` is retired.
2. **All three constructs are sums of separable terms.** Hendrickson's `orig` is exactly a person-level term plus an occasion-level term, each a valid covariance, and is positive definite for every schedule. The current `covar` is an AR(1) term plus an occasion-level term whose factor matrix has zero diagonal and is therefore indefinite. That term carries cross-factor correlation without carrying any variance, which is why `covar` fails on dense schedules. The proposal is a single valid term (Section 4).
3. **`covar` has an exact failure condition.** With a common ρ, it is positive definite if and only if $\omega_{\min}(A)(1-c_\times) > c_1 - c_\times$, where $\omega_{\min}(A)$ is the smallest eigenvalue of the AR(1) kernel. At $\rho = 0.7$ this fails for half-week spacing at the production values $(c_1, c_\times) = (0.2, 0.1)$ and paper 04's $(0.3, 0.2)$, and for quarter-week spacing at $(0.1, 0.05)$ (verified, Section 4.4).
4. **The ceiling on c_{bm} becomes exact and interpretable.** Under separability it factorizes into a factor term and a schedule term, and the schedule term is a sum of per-step costs: on-to-on $(1-\phi)/(1+\phi)$, on-to-off $\phi^2/(1-\phi^2)$, off-to-on $1/(1-\phi^2)$, with $\phi = \rho^{\text{gap}}$ (verified, Section 4.3).
5. **The change reaches most of the compendium.** The same non-separable construct is implemented four times: in the package's `buildSigma()`, in `implementations/tidyverse`, in `implementations/nof1power`, and (in Hendrickson's form) in paper 04's private generator. The response block is shared by both DGP architectures, so mean-moderation results change too. Papers 01-03, 05-11 and 13-14 would need re-runs (Section 7).
6. **Separability matches the analysis model.** Under separability the summed response component is an AR(1) process when the component SDs are constant over visits. With the person-level baseline in the outcome, the data are then a random intercept plus AR(1), which is what the random-intercept-plus-`corCAR1` analysis assumes. Under `covar` they also carry a white-noise term the analysis model does not include (Section 4.6, derived).

2. Review of the original note

The two-page version of 30 September holds up on its central claims, each of which was rechecked numerically for this revision: the positive-definiteness guarantee, the factorized ceiling, the factor term $\sqrt{1-R^2}$ with $R^2 = 0.067$ at $c_1 = 0.2$, and the ceiling table. It had five gaps, addressed below.

- **It did not explain why `covar` fails.** It reported the half-week failure without a mechanism. Section 4.4 derives the exact condition.
- **It did not contrast Hendrickson's construct.** `orig` turns out to be a coherent two-level model, and the comparison clarifies what was lost and gained in moving to `covar` (Sections 3 and 4.5).
- **It understated the scope of the change.** It described an edit to `buildSigma()`. The construct is duplicated across three other engines, affects both architectures, and invalidates the parity baselines (Section 7).
- **It did not note a consequence of the AR(1)-only kernel.** Within a factor, correlation at a 16-week gap is $0.7^{16} = 0.003$ under AR(1), against a constant ρ under `orig` (0.8 as published; 0.7 at the package value). Person-level persistence in the outcome is then carried only by the baseline term BL_i (Section 3).
- **Its main reference was vague.** Section 11 now gives the sources, with their verification status.

3. Intuition

Each participant's symptom score is $Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it})$, and the three components are drawn jointly. The question is how a deviation in one component at one visit relates to deviations in the components at other visits.

Hendrickson's construct: traits plus shocks. `orig` behaves as if each participant carried a fixed offset for each component that lasts the whole trial (a trait), plus an independent weekly disturbance (a shock). Because traits never fade, any two visits of a factor are correlated at ρ, whether one week or twenty weeks apart: 0.8 in the published construct, 0.7 at the package value. The traits are weakly correlated across components and the visit-level shocks more strongly: 0.125 and 0.500 at the published values, 0.143 and 0.333 at $\rho = 0.7$ (verified). This is the classic random-intercept-plus-noise picture, and it is internally consistent.

`covar`: memory that fades, with a leftover. `covar` replaced the everlasting trait with fading memory: today's deviation carries into next week at 0.7, into the week after at 0.49, and so on. That is clinically more plausible. But it kept Hendrickson's two cross-factor numbers. The same-visit cross-factor correlation of 0.2 is Hendrickson's shock-level sharing, and in `orig` it is carried by the shocks' own variance. `covar` has no separate shock variance, so the same-visit excess (0.2 against 0.1 at other visits) is asserted without any variance component to produce it. On widely spaced visits this is harmless. On closely spaced visits, where adjacent measurements of a factor are nearly identical, the matrix is asked to make two nearly identical quantities correlate differently with a third, which no valid covariance can do.

The separable proposal: one clock. The proposal lets all variation flow through a single fading memory shared by the three components. A shared disturbance today affects all three components, and its influence fades at the same rate for each component's own history and for their association with each other. Generatively, the three components are fixed blends of three independent AR(1) processes that tick on the same clock. Nothing is asserted that the clock does not generate, which is why validity cannot fail.

What the AR(1)-only choice gives up. Under fading memory alone the components carry almost no person-level persistence across a 20-week trial. The outcome still has person-level persistence through BL_i, which is a participant-level term in Y_{it}, so the analysis model's random intercept remains supported. If persistent component-level traits are wanted, Section 8 gives the valid extension.

4. Derivations

4.1 Setup and the feasibility condition

For one participant on one path with n visits, the DGP draws $X = (B, \mathrm{BL}, TV_{1:n}, PB_{1:n}, BR_{1:n})$ from a multivariate normal distribution with correlation matrix

$$R = \begin{pmatrix} 1 & 0 & \tilde{r}^\top \\ 0 & 1 & 0^\top \\ \tilde{r} & 0 & M \end{pmatrix}, \qquad \tilde{r} = c_{bm}\,(e_3 \otimes g),$$

where M is the $3n \times 3n$ response block, $e_3 = (0,0,1)^\top$ selects BR, and $g \in \mathbb{R}^n$ is the coupling pattern: $g_t = 1$ on drug, g_t equal to the residual fraction of the on-drug effect off drug, and $g_t = 0$ before any exposure. Variances scale R to the covariance and do not affect positive definiteness. By the Schur complement on the biomarker row, R is positive definite if and only if

$$M \succ 0 \quad \text{and} \quad 1 - \tilde{r}^\top M^{-1} \tilde{r} > 0, \qquad \text{so} \qquad c^*_{bm} = \left((e_3 \otimes g)^\top M^{-1} (e_3 \otimes g)\right)^{-1/2}.$$

The quantity $\tilde{r}^\top M^{-1} \tilde{r}$ is the R^2 of the biomarker regressed on all response variables, so the condition states only that the biomarker cannot be more than perfectly predictable.

4.2 The separable structure

Let $A_{ts} = \rho^{|w_t - w_s|}$ and $K = (1 - c_1) I_3 + c_1 J_3$, and set $M = K \otimes A$, so that $\mathrm{Cor}(X_{c,t}, X_{c',s}) = K_{cc'} A_{ts}$ with factor-major ordering. Three standard Kronecker identities [15] carry the argument, the first for eigenpairs $K q_K = \mu q_K$ and $A q_A = \omega q_A$:

$$(K \otimes A)(q_K \otimes q_A) = (\mu q_K) \otimes (\omega q_A), \qquad (K \otimes A)^{-1} = K^{-1} \otimes A^{-1}, \qquad \det(K \otimes A) = (\det K)^n (\det A)^3.$$

Validity. The eigenvalues of M are the products $\mu_i \omega_j$ of those of K and A. K has eigenvalues $1 + 2c_1$ (once) and $1 - c_1$ (twice), so $K \succ 0$ for $-\frac{1}{2} < c_1 < 1$. A is the correlation matrix of a stationary Ornstein-Uhlenbeck process sampled at the visit times; the exponential kernel $\rho^{|\Delta|} = e^{-|\Delta| \ln(1/\rho)}$ is positive definite for any distinct times [3, 17]. Hence $M \succ 0$ for every schedule.

Generative form. Writing the response as an $n \times 3$ matrix $\mathbf{X}$ with one column per factor, separability is the matrix-normal model $\mathrm{Cov}(\mathrm{vec}\,\mathbf{X}) = K \otimes A$ [8], realized as $\mathbf{X} = L_A \mathbf{Z} L_K^\top$ with $L_A L_A^\top = A$, $L_K L_K^\top = K$ and $\mathbf{Z}$ standard normal. Each column of $L_A \mathbf{Z}$ is an independent AR(1) path; the factors are their fixed mixtures through L_K.

4.3 The factorized ceiling and its switch-cost form

Substituting $M^{-1} = K^{-1} \otimes A^{-1}$ into Section 4.1:

$$(e_3 \otimes g)^\top (K^{-1} \otimes A^{-1})(e_3 \otimes g) = [K^{-1}]_{33}\, g^\top A^{-1} g, \qquad c^*_{bm} = \left([K^{-1}]_{33}\right)^{-1/2} (g^\top A^{-1} g)^{-1/2}.$$

Factor term. For the equicorrelated K, $K^{-1} = (1 - c_1)^{-1}(I_3 - \frac{c_1}{1 + 2c_1} J_3)$, so

$$[K^{-1}]_{33} = \frac{1 + c_1}{(1 - c_1)(1 + 2c_1)} = \frac{1}{1 - R^2}, \qquad R^2 = \frac{2c_1^2}{1 + c_1},$$

where R^2 is the squared multiple correlation of BR on TV and PB at one visit: 0.0182, 0.0667 and 0.1385 at $c_1 = 0.1, 0.2, 0.3$ (verified).

Schedule term. A sampled Ornstein-Uhlenbeck process is Markov: $X_t = \phi_t X_{t-1} + \sqrt{1 - \phi_t^2}\, \xi_t$, with ξ_t independent standard normal innovations, $\phi_t = \rho^{d_t}$ and $d_t = w_t - w_{t-1}$ [3]. Factorizing the joint density gives $A^{-1} = L^\top L$ with L lower bidiagonal, $L_{11} = 1$, $L_{tt} = (1 - \phi_t^2)^{-1/2}$ and $L_{t,t-1} = -\phi_t (1 - \phi_t^2)^{-1/2}$, so

$$g^\top A^{-1} g = g_1^2 + \sum_{t=2}^{n} \frac{(g_t - \phi_t g_{t-1})^2}{1 - \phi_t^2}.$$

Each term is the squared, standardized one-step prediction error of the coupling pattern under the AR(1) predictor. For a binary pattern the steps cost:

Table 3: Cost of each coupling-pattern step under the AR(1) predictor

Step	Cost	Short gap ($\phi \to 1$)
on to on	$(1-\phi)/(1+\phi)$	small
on to off	$\phi^2/(1-\phi^2)$	large
off to on	$1/(1-\phi^2)$	large
off to off	0	none

The ceiling is therefore $c_{bm}^{*-2} = [K^{-1}]_{33}\{g_1^2 + \sum \text{step costs}\}$: a coupling pattern that changes abruptly between closely spaced visits is expensive, and one that is constant across closely spaced visits is cheap. The formula agrees with the full matrix to 9×10^{-16} on the paper 01 binding paths and a 16-visit multi-cycle schedule (verified). It also corrects the approximation in `docs/34-cbm-feasible-range.md` Section 5.5, which assigned every switch the off-to-on cost; an on-to-off switch costs exactly one unit less. A pattern that starts on drug pays $g_1^2 = 1$ up front, so the two orderings of a single switch cost the same in total.

For a constant pattern ($g = 1$), $g^\top A^{-1} g = 1 + \sum_t (1-\phi_t)/(1+\phi_t) = n_{\text{eff}}$, the effective number of independent visits, and the ceiling is $\sqrt{(1-R^2)/n_{\text{eff}}}$.

4.4 Why the current `covar` fails

With a common ρ, `covar` sets within-factor entries to A_{ts}, same-visit cross-factor entries to c_1, and different-visit cross-factor entries to $c_\times A_{ts}$. Hence (verified, exact reconstruction):

$$M_{\text{covar}} = K_a \otimes A + K_b \otimes I_n, \qquad K_a = (1-c_\times) I_3 + c_\times J_3, \qquad K_b = (c_1 - c_\times)(J_3 - I_3).$$

K_b has zero diagonal and positive off-diagonal entries, so it is indefinite, with eigenvalues $2(c_1 - c_\times)$ and $-(c_1 - c_\times)$ (twice). It is the same-visit excess correlation, attached to no variance. Diagonalizing $A = Q\Omega Q^\top$ with $\Omega = \text{diag}(\omega_1, \ldots, \omega_n)$, $(I_3 \otimes Q)^\top M_{\text{covar}} (I_3 \otimes Q) = K_a \otimes \Omega + K_b \otimes I_n$, which a permutation of rows and columns (factor-major to visit-major) turns into the block-diagonal

$$\bigoplus_{j=1}^{n} (\omega_j K_a + K_b),$$

and since K_a and K_b share eigenvectors, the eigenvalues of M_{covar} are $\omega_j(1 + 2c_\times) + 2(c_1 - c_\times)$ and $\omega_j(1 - c_\times) - (c_1 - c_\times)$. Therefore

$$M_{\text{covar}} \succ 0 \iff \omega_{\min}(A) > \frac{c_1 - c_\times}{1 - c_\times}.$$

The predicted smallest eigenvalue matches the computed one to 10^{-15} on six schedules and three parameter sets (verified). For equally spaced visits with $\phi = \rho^d$, the eigenvalues of the AR(1) Toeplitz matrix lie above the minimum of its spectral density, $(1-\phi)/(1+\phi)$, and approach it as n grows [16]. A sufficient condition for validity at any n is then $(1-\phi)/(1+\phi) \geq (c_1 - c_\times)/(1 - c_\times)$, which at $\rho = 0.7$ requires gaps of at least 0.63 weeks for $(0.2, 0.1)$, 0.70 weeks for $(0.3, 0.2)$, and 0.30 weeks for $(0.1, 0.05)$ (corrected 2026-10-08 from 0.71: the condition gives $\ln(7/9)/\ln 0.7 = 0.7046$; see `analysis/report/compendium-mathematics.tex`, Section "The `covar` spectrum, a spacing condition"). Every production design examined satisfies this; dense or long schedules may not.

4.5 Hendrickson's construct as two valid terms

`orig` sets within-factor entries to ρ at every lag and cross-factor entries to c_1 (same visit) and $c_\times$ (otherwise). Hence (verified, exact reconstruction):

$$M_{\text{orig}} = K_o \otimes I_n + K_p \otimes J_n, \qquad K_p = \rho I_3 + c_\times(J_3 - I_3), \qquad K_o = (1-\rho)I_3 + (c_1 - c_\times)(J_3 - I_3).$$

This is the covariance of $X_{c,t} = T_c + O_{c,t}$, with person-level effects $T \sim N(0, K_p)$ and occasion-level effects $O_t \sim N(0, K_o)$ independent across visits [2]. The implied cross-factor correlations are $c_\times/\rho$ at the person level and $(c_1 - c_\times)/(1-\rho)$ at the occasion level: 0.143 and 0.333 at the package value $\rho = 0.7$, and 0.125 and 0.500 at the published $\rho = 0.8$ (verified). Because J_n has eigenvalues n and 0, $M_{\text{orig}} \succ 0$ if and only if $K_o \succ 0$ and $K_o + nK_p \succ 0$; with $K_p \succeq 0$ this reduces to $\mu_{\min}(K_o) = (1-\rho) - (c_1 - c_\times) > 0$, independent of n and the schedule. It is 0.20 at $\rho = 0.7$ (verified for $n = 4$ to 32) and 0.10 at the published $\rho = 0.8$ (verified). This recovers the expression given in paper 01 Appendix B.7.

`orig` is a single separable term only when $K_o \propto K_p$, that is when $c_\times = c_1\rho$, in which case $M_{\text{orig}} = K \otimes A_{\text{CS}}$ (verified: the difference is 0.040 at $c_\times = 0.1$ and 0 at $c_\times = 0.14$).

4.6 The summed response and the analysis model

The analysis model regresses Y_{it}, which contains the sum of the three components, with a random intercept and `corCAR1` residuals. Let $s_t = (s_{TV,t}, s_{PB,t}, s_{BR,t})^\top$ hold the factor standard deviations at visit t and $S_t = \sum_c s_{c,t} X_{c,t}$ the summed component. Under the separable structure,

$$\text{Cov}(S_t, S_s) = (s_t^\top K s_s)\, A_{ts},$$

which is exactly proportional to the AR(1) kernel whenever the factor standard deviations are constant over visits. In this construct the PB SD is 10 times the visit's expectancy weight (1 open label, 0.5 blinded), so it differs between open-label and blinded visits. The correlation factor $s_t^\top K s_s / \sqrt{(s_t^\top K s_t)(s_s^\top K s_s)}$ between an open-label and a blinded visit is 0.976 (verified), so the departure from AR(1) is small.

The outcome is $Y_{it} = \text{BL}_i - S_{it}$, and BL_i is a person-level constant independent of the responses. Hence $\text{Cov}(Y_{it}, Y_{is}) = \sigma^2_{BL} + \text{Cov}(S_t, S_s)$: a random intercept plus an AR(1) process. That is exactly the residual structure of the random-intercept-plus-`corCAR1` analysis model [5], not of `corCAR1` alone. Under `covar`, by the decomposition of Section 4.4,

$$\text{Cov}(S_t, S_s) = (s_t^\top K_a s_s)\, A_{ts} + (s_t^\top K_b s_t)\, \mathbf{1}[t = s],$$

an AR(1) process plus a white-noise term, since $s^\top K_b s = 2(c_1 - c_\times)(s_{TV}s_{PB} + s_{TV}s_{BR} + s_{PB}s_{BR}) > 0$. A `corCAR1` residual model does not include such a term, so under `covar` the analysis model's residual structure is misspecified even before the biomarker enters, whereas under separability it is correctly specified up to the time-varying placebo-belief scale (which the package ties to the expectancy weight). Both statements follow algebraically from the verified decompositions; their effect on test calibration has not been measured.

4.7 One framework for all three

All three constructs are instances of the linear model of coregionalization, $M = \sum_j K_j \otimes A_j$, in which each term is a factor matrix times a correlation kernel [13, 14]. A sum of such terms is guaranteed valid when every K_j is positive semidefinite.

Table 4: Comparison of the `orig`, `covar` and proposed response covariance constructs

	`orig` (58b32a9)	`covar` (current)	Proposed
Terms	$K_p \otimes J + K_o \otimes I$	$K_a \otimes A_{\mathrm{AR}} + K_b \otimes I$	$K \otimes A_{\mathrm{AR}}$
Factor matrices	both PSD	K_b indefinite	PSD
Interpretation	person traits + occasion shocks	fading memory + unbacked same-visit excess	fading memory shared by all factors
Within-factor correlation at 16 weeks	ρ (0.8 published, 0.7 package)	0.003	0.003
Valid for every schedule	yes	no (Section 4.4)	yes
Ceiling depends on	visit count and on-share only (binary coupling)	full schedule	full schedule, in closed form
Cross-factor parameters	c_1, $c_\times$	c_1, $c_\times$	c_1

4.8 The ceiling with prognostic and baseline couplings

The stress test of `docs/46` couples the biomarker to the natural course, the placebo response or the baseline level as well as to the drug response. Write the biomarker's correlations with the three response factors as vectors v_{TV}, v_{PB}, v_{BR} over the visits, and its correlation with BL as c_{bl}. Since BL is uncorrelated with the responses, the Schur complement of Section 4.1 gives

$$R \succ 0 \iff M \succ 0 \quad \text{and} \quad c_{bl}^2 + \sum_{c,c'} [K^{-1}]_{cc'}\, v_c^\top A^{-1} v_{c'} < 1,$$

using $M^{-1} = K^{-1} \otimes A^{-1}$. The quadratic form agrees with the full matrix to 10 digits on the Hybrid design (verified, `analysis/scripts/quick-sim/cbm-ceiling/06-separable-review-checks.R`). Here $v_{BR} = c_{bm}\, g$, and a constant prognostic coupling to the natural course is $v_{TV} = c_{bm,TV}\mathbf{1}$. With $c_{bm} = c_{bm,TV} = c_{bl} = 0.3$ on Hybrid path A, the left side is 0.75 (verified, 0.7525), and the stress-test cells are feasible.

The off-diagonal elements of K^{-1} are negative, $-c_1/((1-c_1)(1+2c_1))$. Couplings of the same sign to correlated factors therefore cost less than the sum of their separate costs. On Hybrid path A with $c_1 = 0.2$, coupling 0.3 to BR alone costs 0.390, 0.3 to TV alone costs 0.342, and both together cost 0.663: the cross term is -0.070 (verified). A biomarker that is both predictive and prognostic in the same direction is accordingly easier to accommodate than either coupling alone would suggest.

5. Advantages and disadvantages

Advantages.

1. **Validity for every schedule.** No spacing, length or irregularity can make the response block indefinite (Section 4.2).
2. **No implementation ambiguity.** There is no per-pair cross-factor decay rate, so the loop-order dependence in `buildSigma()` and its copies cannot arise.
3. **An exact, interpretable ceiling** that can be checked for every simulation cell before data are drawn (Section 4.3).
4. **A recognized structure.** Kronecker-product covariances are the standard model for multivariate repeated measures [6, 7]; estimation [8] and tests of separability [9, 10] are established.
5. **Fewer parameters.** One cross-factor parameter instead of two.
6. **Faster computation.** Inversion and Cholesky factorization reduce to the 3×3 and $n \times n$ factors.
7. **Coherence with the analysis model.** With constant component SDs, the outcome is a random intercept plus AR(1), the structure of the random-intercept-plus-`corCAR1` analysis. Under `covar` it carries an additional white-noise term (Section 4.6).

Disadvantages.

1. **Results change.** Lagged cross-factor correlations double at production values ($c_1\rho^{\text{gap}}$ instead of $c_\times\rho^{\text{gap}}$ with $c_\times = c_1/2$). No current result reproduces exactly.
2. **One persistence for all factors.** Factor-specific ρ (for example faster-fading placebo belief) is excluded. The valid extension is a sum of separable terms (Section 8), at the cost of the closed-form ceiling.
3. **No same-visit excess.** The ratio of cross-factor to within-factor correlation is c_1 at every lag, so cross-factor association cannot be stronger at the same visit than the within-factor pattern implies. If occasion-level shared shocks are substantively important, a valid occasion-level term must be added explicitly (Section 8).
4. **No lead-lag structure.** One factor cannot predict another at a later visit more strongly than the reverse; neither current construct allows this either.
5. **Separability is a testable assumption** [9, 10], and it is frequently rejected in applied multivariate and space-time data, which has motivated nonseparable alternatives [11, 12]. Whether symptom-component data would reject it is unknown, since the components are latent. For a simulation DGP this is a modeling commitment to state, not a defect.

6. Effect on the largest testable c_{bm}

Ceilings without carryover at $\rho = 0.7$, $c_1 = 0.2$ (current structure also $c_\times = 0.1$); closed form and full matrix agree to 10^{-16} (verified):

Table 5: Ceilings without carryover under the current and separable AR(1) structures by path

Path	Current `covar`	Separable AR(1)
CO, path A	0.619	0.616
Hybrid, path A	0.456	0.480
Hybrid, path C	0.465	0.491
OL+BDC, path A	0.452	0.474
Multi-cycle, 16 weekly visits	0.262	0.286
Dense, 0.5-week spacing	not positive definite	0.509

At Hendrickson et al.'s published $\rho = 0.8$ the separable AR(1) ceiling for the Hybrid design without carryover is lower, 0.440 (`docs/36`, Section 6.5; configuration C). For comparison, the compound-symmetry ceiling of `orig` without carryover is 0.256 (`docs/36`, Section 4.2).

The design conclusions of `docs/34` carry over unchanged in direction, because they are driven by the schedule term, which separability makes exact rather than approximate. Their numbers would need recomputing.

A higher ceiling is not more power. At the same c_{bm}, AR(1) data carry much less information about the interaction than compound-symmetric data. The person-level persistence that compound symmetry provides cancels in a within-participant contrast; fading memory does not. In the Hybrid design at $c_{bm} = 0.25$, the closed-form power of the paired-difference statistic is 0.721 under compound symmetry and 0.335 under the separable structure (`docs/37`, Section 5.8). Effect sizes under the two structures should therefore be compared on the scale of the implied interaction slope, not of c_{bm}.

Graded coupling and the step costs. Under graded coupling the off-drug pattern is not binary: $g_t = 2^{-t_{sd,t}/t_{1/2}}$. The general term $(g_t - \phi_t g_{t-1})^2/(1-\phi_t^2)$ of Section 4.3 then replaces the binary on-to-off cost $\phi_t^2/(1-\phi_t^2)$ with $(g_t - \phi_t)^2/(1-\phi_t^2)$, which is smaller when g_t is close to ϕ_t. This is why the ceiling under graded coupling rises with the half-life.

7. Impact across the compendium

The response block is shared by both architectures: mean moderation draws TV, PB and BR from the same matrix and adds its shift afterwards. The change therefore affects every paper whose data come from one of the engines below, regardless of architecture.

Table 6: Simulation engines with their response covariance construct and parameters

Engine	Construct	Parameters
`R/generateData.R` (`buildSigma`)	`covar`	set by drivers
`implementations/tidyverse`	`covar`, same loops	set by drivers
`implementations/nof1power`	`covar`, same loops	set by drivers
Paper 04 private generator (`treatment-main-effect/vig*.R`)	`orig`-like (CS within, constant cross) with the 2024 scale factor	$(0.3, 0.2)$

Per paper, from the drivers (inspected; production designs were not all re-checked against the validity condition of Section 4.4):

Table 7: Engine, cross-factor parameters and consequence of the change for each paper

Paper	Engine	$(c_1, c_\times)$	Consequence
01 DGP architectures	package	(0.2, 0.1)	Section 3 re-run already required; Appendix B.3, B.5.2, B.6-B.7 and the ceiling discussion rewritten around one construct
02 Carryover sensitivity	tidyverse	(0.2, 0.1)	Full re-run; decay-shape study should also decay the correlation with the chosen shape
03 Latent class	package	(0.1, 0.05)	Re-run of mixture and comparator DGPs
04 Main effect	private, `orig`-like	(0.3, 0.2)	Unaffected unless migrated; decide whether paper 04 should adopt the unified DGP
05 Design sensitivity	nof1power	(0.1, 0.05)	Re-run; design sweeps over period length should be checked against Section 4.4, since short periods are where `covar` fails
06 Component decomposition	package	(0.1, 0.05)	Re-run; its contaminated-biomarker variant couples B to PB as well, which generalizes the ceiling to $\sum_{cc'}[K^{-1}]_{cc'}\, v_c^\top A^{-1} v_{c'}$ (Section 4.8)
07 Gompertz evaluation	package	(0.2, 0.1)	Re-run

Paper	Engine	$(c_1, c_\times)$	Consequence
08 Test procedure and design	package	(0.1, 0.05)	Re-run; cycle and period sweeps to be checked against Section 4.4
09 Informative dropout	package	(0.1, 0.05)	Re-run; its reproduction of Hendrickson's Figure 4A moves further from `orig`
10 Test calibration	tidyverse (via paper 02 core)	(0.2, 0.1)	Re-run; standard-error calibration depends directly on the residual covariance, and part of the `corCAR1` miscalibration it reports may come from the white-noise term of Section 4.6 (not tested)
11 Combined architecture	package	driver-set	Re-run
12 Design efficiency	closed form, single-factor AR(1)	none	Unaffected; its closed forms extend exactly under separability by the factor term
13, 14 Two-component	package, `components`	(0.2, 0.1)	Re-run; with two factors $[K^{-1}]_{22} = 1/(1 - c_1^2)$

Two infrastructure items also change. The cross-implementation parity baselines (`analysis/scripts/parity/`) would need regenerating, and the Hendrickson comparison arm should be rebased on the `58b32a9` sources, since the vendored copy now in the repository is the 2024 state.

8. Alternatives considered

- **Keep `covar` and fix the loop order.** Retains the failure region of Section 4.4 and the unbacked same-visit excess.
- **Separable AR(1) (proposed).** One valid term, one clock.
- **Separable AR(1) plus an occasion-level term,** $K_s \otimes A_{\mathrm{AR}} + K_o \otimes I$ with $K_o \succeq 0$. This legitimately restores a same-visit excess, at the price of a within-factor nugget (within-factor correlation below 1 at lag zero-plus) and the loss of the closed-form ceiling.
- **Person-level term plus AR(1),** $K_p \otimes J + K_s \otimes A_{\mathrm{AR}}$. Restores Hendrickson's persistent traits within a fading-memory model, the decomposition advocated by Diggle [2, 4]. The AR(1)-only kernel was chosen instead; this remains the natural route if component-level persistence is later wanted.
- **Separable compound symmetry,** $K \otimes A_{\mathrm{CS}}$: Hendrickson's construct with $c_\times = c_1\rho$. Valid and simple, but without decay with lag.

9. Recommendation and implementation

Adopt the separable AR(1) structure for the single covariance-moderation DGP, and use the same response block for mean moderation. Implementation steps:

1. In `buildSigma()`, replace the within- and cross-factor loops with `kronecker(K, A)` over the retained components, keeping factor-major labels; retire `c.cfct` (error if supplied, rather than ignore).

2. Add `cbmCeiling()` implementing Section 4.3, and call it before simulation in `generateSimulatedResults()` and `validateParameterGrid()`, stopping when $c_{bm} \geq c^*_{bm}$. Remove the silent positive-definiteness repair for this construct.
3. Apply the same change to `implementations/tidyverse` and `implementations/nof1power`, or retire them in favor of the package.
4. Tests: positive definiteness at quarter-week spacing; closed-form ceiling equal to the numerical ceiling; exact equality with the current construct when $c_1 = c_\times = 0$, where the two coincide; two-component case.
5. Regenerate parity baselines and re-run the affected papers in the order of Section 7.

10. Evidence status and limitations

- **Verified by computation** (`analysis/scripts/quick-sim/cbm-ceiling/ 05-separable-derivations.R`, run 2026-10-01): the decompositions of `covar` and `orig` (exact reconstruction), the validity condition for `covar` (eigenvalues to 10^{-15}), the closed form of $[K^{-1}]_{33}$, the switch-cost form of $g^\top A^{-1} g$, the separability condition for `orig`, and the ceilings of Section 6.

- **Derived, not separately computed:** the equal-spacing gap thresholds of Section 4.4, which rest on the Toeplitz eigenvalue bound [16] and are consistent with the computed eigenvalues; and the covariance of the summed response in Section 4.6, which follows from the verified decompositions. Its effect on paper 10's calibration results is a hypothesis.

- **Inspected:** the engines and parameter values of Section 7, from the driver sources. Paper 04's production engine was not traced in full, and the production designs of papers 05 and 08 were not checked against the validity condition.

- **Simulated since 2026-10-01.** The separable structure at $\rho = 0.7$ (configuration C of `docs/36`) has been used in three studies, with no positive-definiteness failure at c_{bm} up to 0.45:

  – a power arm of `docs/36`;
  – the configuration C arm (C07) of the covariance study (`docs/45`, 378,000 fits);
  – the configuration C cells of the stress test (`docs/46`, script 11: 41 cells, the 39 of the original protocol plus two prognostic-baseline cells, 15 null and 26 alternative, 28,000 trials in all, including prognostic and baseline couplings).

  Configuration C is the covariance construct of the refined process (paper 01), but neither study ran the refined process in full: both kept the published recursive carryover and the fixed published allocation (inspected in `08-covariance-study.R` and `11-strawman-stress-test.R`).

  Under these data (corrected 2026-10-07; the earlier wording said "nominal everywhere", which holds only without prognostic coupling):

  – the random-intercept-plus-`corCAR1` analyses with CR2 standard errors (S and its variants C1, DS, ES, F1, F2, G1) had pooled null rejection 0.048 to 0.053 over the 11 non-prognostic null cells, and the model-based variants with phase-specific variances (C2, G2) 0.047 to 0.052 (verified, `decision-screen-ds-nosym-all14.csv`);
  – under prognostic coupling several of them were anticonservative (S 0.100 with a placebo-prognostic biomarker; C1 and C2 0.38 to 0.39 with a constant natural-course association), whereas ES, EA, F1, G1 and G2 stayed near nominal. This is a mean-model failure, not a covariance one; the decision rule rerun with all 14 analyses selects F1 (`docs/46`, Amendment 1; `docs/49`);
  – a compound-symmetry working model was anticonservative in the crossover design, with $\kappa = 0.74$ derived in `docs/45`, Section 4.3.

  The non-prognostic results are consistent with Section 4.6, but they do not test the white-noise hypothesis about `covar`.

- **Adopted** as the reference structure in paper 01 (Methods, "Why separability"; Appendix A.5).

- **Not implemented in the package.** Steps 1 to 3 of Section 9 are pending. The package's sampler still repairs a matrix silently when its Cholesky factorization fails (`R/generateData.R`, lines 391 and 392; inspected 2026-10-07). The structure exists in the simulation drivers only (`04-power-simulation.R` arm C; `08-covariance-study.R`; `11-strawman-stress-test.R`).
- **References.** All 17 were checked on 2026-10-07 (Section 11).

11. References

Status (2026-10-07): [2] and [3] verified in PubMed (PMIDs 3233259 and 2049497); [1], [2], [3], [6] to [13] verified against Crossref (authors, year, journal, volume, issue, pages; DOIs below); the books [4], [5], [14], [15] and [17] matched to their publisher records in Crossref; [16] cited from the standard bibliographic record (no DOI).

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2020;2:13. doi:10.3389/fdgth.2020.00013.
2. Diggle PJ. An approach to the analysis of repeated measurements. *Biometrics* 1988;44(4):959-971. doi:10.2307/2531727.
3. Jones RH, Boadi-Boateng F. Unequally spaced longitudinal data with AR(1) serial correlation. *Biometrics* 1991;47(1):161-175. doi:10.2307/2532504.
4. Diggle PJ, Heagerty P, Liang K-Y, Zeger SL. *Analysis of Longitudinal Data*, 2nd ed. Oxford University Press; 2002. doi:10.1093/oso/9780198524847.001.0001.
5. Pinheiro JC, Bates DM. *Mixed-Effects Models in S and S-PLUS*. Springer; 2000. doi:10.1007/b98882.
6. Galecki AT. General class of covariance structures for two or more repeated factors in longitudinal data analysis. *Communications in Statistics: Theory and Methods* 1994;23(11):3105-3119. doi:10.1080/03610929408831436.
7. Naik DN, Rao SS. Analysis of multivariate repeated measures data with a Kronecker product structured covariance matrix. *Journal of Applied Statistics* 2001;28(1):91-105. doi:10.1080/02664760120011626.
8. Dutilleul P. The MLE algorithm for the matrix normal distribution. *Journal of Statistical Computation and Simulation* 1999;64(2):105-123. doi:10.1080/00949659908811970.
9. Lu N, Zimmerman DL. The likelihood ratio test for a separable covariance matrix. *Statistics and Probability Letters* 2005;73(4):449-457. doi:10.1016/j.spl.2005.04.020.
10. Mitchell MW, Genton MG, Gumpertz ML. A likelihood ratio test for separability of covariances. *Journal of Multivariate Analysis* 2006;97(5):1025-1043. doi:10.1016/j.jmva.2005.07.005.
11. Genton MG. Separable approximations of space-time covariance matrices. *Environmetrics* 2007;18(7):681-695. doi:10.1002/env.854.
12. Gneiting T. Nonseparable, stationary covariance functions for space-time data. *Journal of the American Statistical Association* 2002;97(458):590-600. doi:10.1198/016214502760047113.
13. Goulard M, Voltz M. Linear coregionalization model: tools for estimation and choice of cross-variogram matrix. *Mathematical Geology* 1992;24(3):269-286. doi:10.1007/BF00893750.
14. Wackernagel H. *Multivariate Geostatistics*, 3rd ed. Springer; 2003. doi:10.1007/978-3-662-05294-5.
15. Horn RA, Johnson CR. *Topics in Matrix Analysis*. Cambridge University Press; 1991. doi:10.1017/CBO9780511840371.
16. Grenander U, Szegő G. *Toeplitz Forms and Their Applications*. University of California Press; 1958.
17. Rasmussen CE, Williams CKI. *Gaussian Processes for Machine Learning*. MIT Press; 2006. doi:10.7551/mitpress/3206.001.0001.